

Select and use equipment to generate and record data with precision to obtain useful sample sizes and replicable data, using digital tools as appropriate

Learning intention

- select and use equipment to generate and record data with precision to obtain useful sample sizes and replicable data, using digital tools as appropriate.

Teach and model

- deciding how much data is needed to produce valid conclusions.
- identifying how human error can affect replicability and reproducibility.
- explaining how estimation affects precision and examining the inaccuracy introduced when reading between scale markings.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.